WORKED SOLUTION

STEP II 2004, Question 9

Mechanics • Centre of Gravity

STEP II 2004, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s2-q9

The key idea

A hemisphere resting on its curved surface is touching the plane at a point of a sphere, and the normal to a sphere always passes through its centre. So the reaction acts along the vertical through O, the centre of the base, and in equilibrium the centre of gravity of whatever is resting there must lie on that same vertical line. Each part of the question is then just "find the centre of gravity, and tilt until it is directly below O". Coordinates fixed in the body make this painless.

Solution

The model and the equilibrium condition

Treat the hemisphere as a rigid body touching a horizontal plane at a single point C, and the particle as a point mass fixed to the rim. The only forces are the weights and the contact force at C. Since C is on the spherical surface, the normal at C passes through O, and since the plane is horizontal this normal is vertical. So C is directly below O, and O stays at height r however the body rolls.

In equilibrium the contact force must balance the total weight, so it is vertical (there is no friction) and acts along the same line as the weight. Hence:

in equilibrium, the centre of gravity of the system lies on the vertical line through O.

It lies below O, because the body sits on the curved side of the base.

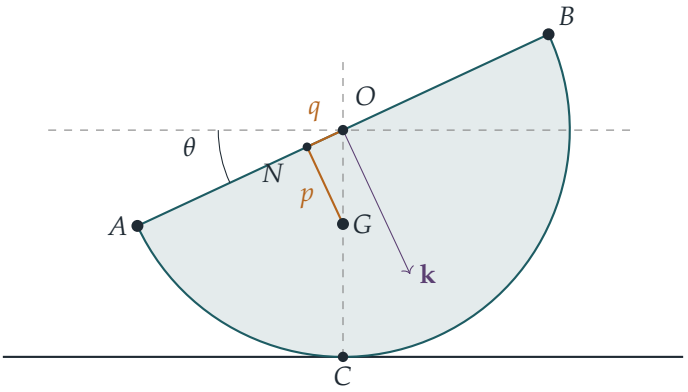
Body coordinates and the initial position

Fix two perpendicular unit vectors in the body: $\mathbf{k}$ along the axis of symmetry, from O towards the curved surface, and $\mathbf{e}$ in the plane of the base. Since G is a distance p from the base and $\sqrt{p^2+q^2}$ from O, its foot N on the base is a distance q from O. Choosing

e to point from O towards N, we get

$$\overrightarrow{OG} = p\,\mathbf{k} + q\,\mathbf{e}.$$

The base is perpendicular to **k**, so the angle the base makes with the horizontal equals the angle between **k** and the vertical. In the initial position $\overrightarrow{OG}$ is vertically downwards, so this angle θ satisfies $\tan\theta = q/p$.



Which rim point is lowest? A point on the rim is $r(\cos\phi\,\mathbf{e} + \sin\phi\,\mathbf{f})$ from O, where **f** is the third unit vector, perpendicular to both. Its depth below O is its component along the downward unit vector $(p\mathbf{k} + q\mathbf{e})/\sqrt{p^2+q^2}$, namely

$$\frac{rq\cos\phi}{\sqrt{p^2+q^2}},$$

which is greatest when $\phi = 0$ (taking $q > 0$; if $q = 0$ the base is horizontal and there is no single lowest point). So

$$\overrightarrow{OA} = r\,\mathbf{e}, \qquad \overrightarrow{OB} = -r\,\mathbf{e}.$$

In words: A is on the same side of the axis as G, in the vertical plane through O and G, and the whole problem stays in that plane.

The particle at A

The centre of gravity G_A of hemisphere and particle is given by

$$(M+m)\,\overrightarrow{OG_A} = M(p\,\mathbf{k} + q\,\mathbf{e}) + mr\,\mathbf{e} = Mp\,\mathbf{k} + (Mq + mr)\,\mathbf{e}.$$

In the new equilibrium $\overrightarrow{OG_A}$ is vertical, so, exactly as for θ, the angle between **k** and the vertical (which is the angle between the base and the horizontal) is the angle between $\overrightarrow{OG_A}$ and **k**:

$$\tan\alpha = \frac{Mq + mr}{Mp}, \qquad \text{so} \qquad \alpha = \tan^{-1}\left(\frac{Mq + mr}{Mp}\right).$$

As we would expect, the extra weight at A pulls A further down, and $\alpha > \theta$.

Ask yourself Why is the base angle the angle between $\overrightarrow{OG_A}$ and the axis?

The base is perpendicular to the axis $\mathbf{k}$, and the horizontal is perpendicular to the vertical. Rotating both lines of a pair through a right angle doesn't change the angle between them, so the angle between base and horizontal equals the angle between $\mathbf{k}$ and the vertical. In equilibrium the vertical through O is the line OG_A, whose components along $\mathbf{k}$ and $\mathbf{e}$ are Mp and $Mq + mr$ (after dividing by $M + m$). The tangent of the angle it makes with $\mathbf{k}$ is the ratio of these.

The particle at B

Now the particle is at $-r\,\mathbf{e}$, so the new centre of gravity G_B satisfies

$$(M+m)\,\overrightarrow{OG_B} = Mp\,\mathbf{k} + (Mq - mr)\,\mathbf{e}, \qquad \text{giving} \qquad \tan\beta = \frac{Mq - mr}{Mp}.$$

Because m is small we have $mr < Mq$, so $\overrightarrow{OG_B}$ is still on the same side of the axis as before and the base tilts the same way, through the smaller angle β.

The angle between the two positions

By the compound-angle formula,

$$\begin{aligned}\tan(\alpha-\beta) &= \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta} = \frac{\dfrac{2mr}{Mp}}{1 + \dfrac{(Mq+mr)(Mq-mr)}{M^2p^2}} \\ &= \frac{2mMrp}{M^2p^2 + M^2q^2 - m^2r^2} = \frac{2mMrp}{M^2\,(p^2+q^2) - m^2r^2},\end{aligned}$$

multiplying top and bottom by M^2p^2 in the third step. This is the required result.

Checks

- If $q = 0$ (for instance a uniform hemisphere) the two positions are mirror images: $\tan\alpha = mr/(Mp)$ and $\beta = -\alpha$. The formula gives $\tan 2\alpha = 2t/(1-t^2)$ with $t = mr/(Mp)$, the double-angle formula, as it should.
- If $m \to 0$, then α and β both tend to θ and the right-hand side tends to 0.
- With $M = 1$, $m = 0.1$, $r = 1$, $p = \frac{3}{8}$, $q = \frac{1}{8}$ we get $\tan\alpha = 0.6$ and $\tan\beta = \frac{1}{15}$, so $\alpha \approx 30.96°$ and $\beta \approx 3.81°$. Both sides of the result equal $\frac{20}{39} \approx 0.5128$. We also found the equilibrium angles numerically by minimising the height of the centre of gravity for 2000 random sets of values, and they agreed with the formulae for $\tan\alpha$ and $\tan\beta$ to within 10^{-13}.

Common slips

- Putting G vertically below the *contact point of the original position* rather than below O. The contact point moves as the body rolls; O is the point that is always directly above it.
- Taking A on the wrong side of the axis. The lowest rim point is on the same side as G, which is why the particle at A *increases* the tilt.
- Treating β as positive when $mr > Mq$. Then the base tilts the other way and the formula holds only with β counted as negative; this is what the condition that m is small rules out.

Beyond STEP

The principle used here is that a rigid body in equilibrium under gravity and a single contact force has its centre of mass on the line of that force. Equivalently, the centre of mass is at a stationary height, and checking that height as the body rolls also decides whether the equilibrium is stable. That is the start of the energy method for stability that first-year mechanics develops for rocking and rolling bodies. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers centres of mass and rigid-body equilibrium in this spirit.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.